

NMF vs PCA: Why Non-Negative Matrix Factorization Enhances Interpretability

Panel A: PCA Components
(Orthogonal, can be negative)

$PC1 = +0.5 \times \text{Showers} - 0.3 \times \text{Baths} + 0.2 \times \text{Garden}$

$PC2 = -0.4 \times \text{Showers} + 0.6 \times \text{Baths} - 0.1 \times \text{Leaks}$

⚠ Interpretation Issues:

- Negative loadings: "High showers MINUS baths"
- Difficult to explain as behavioral patterns

Panel B: NMF Components
(Non-negative, additive parts)

$H1 = +0.7 \times \text{Showers} + 0.4 \times \text{Baths} + 0.0 \times \text{Garden}$

"Indoor Bathing Intensity"

$H2 = +0.0 \times \text{Showers} + 0.0 \times \text{Baths} + 0.6 \times \text{Leaks}$

"Infrastructure Quality"

✅ Interpretation Benefits:

- All loadings ≥ 0 : No subtraction needed
- Intuitive "behavioral pattern" interpretation

Panel C: Household Representation
(Additive combination)

$\text{Household } i = w_1 \times H_1 + w_2 \times H_2$

(Purely additive: no negative weights)

Example: High-Intensity Household

$w_1 = 0.9$ (high indoor intensity)

$w_2 = 0.7$ (poor infrastructure)



→ Intervention: Target behavior + fix leaks